

ASSIGNMENT (Aug 18, 2026)
RNA-Seq Literature Familiarization and Data Characterization

Preparation for RNA-Seq Re-analysis Using Galaxy and GitHub

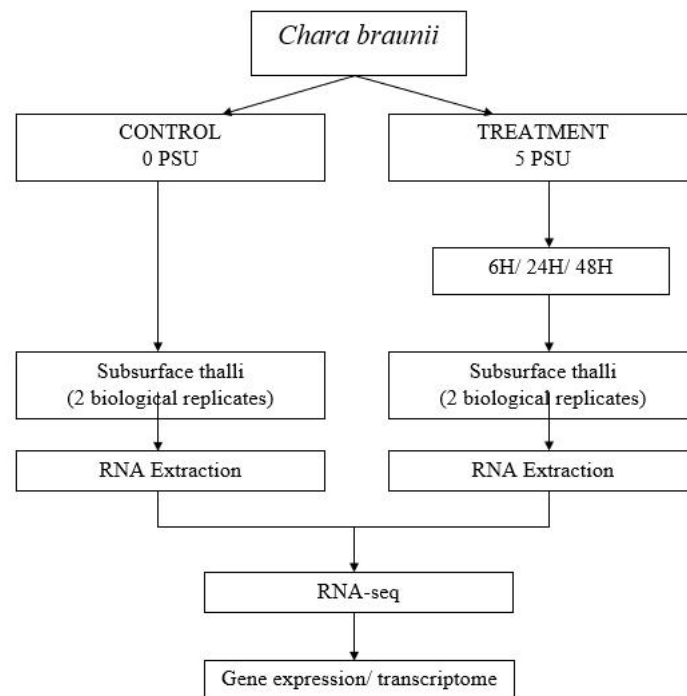
Name	Catipay, Niña Karylle A.
Course and Section	BIO 300 - A
Date	August 18, 2026
Instructors	Ma'am Lilibeth Bucol & Sir Abner Bucol
Group Topics	Salinity Stress

Part 3. Understand the Paper Before Downloading Data

Information from the Paper	Group Answer
Title of the paper	The impact of salt stress on the physiology and the transcriptome of the model streptophyte green alga <i>Chara braunii</i> .
Authors	Daniel Heß, Carolin M. Heise, Hendrik Schubert, Wolfgang R. Hess, Martin Hagemann
Year	2023
Journal	Physiologia Plantarum, 175(6), e14123
Organism	<i>Chara braunii</i>
Tissue / organ / cell type	Subsurface thalli segments
Biological question	How does salt stress affect the physiology and gene expression of <i>Chara braunii</i> ?
Control condition	0 practical salt units (PSU)/ freshwater
Treatment or stress condition	5 PSU
Number of biological replicates	7 replicates
Sequencing platform	Illumina

Single-end or paired-end sequencing	paired-end sequenced
Read length, if reported	2x100 bp
RNA-seq repository	Sequence Read Archive
BioProject / Study accession	PRJNA965918
Run accession numbers	SRR24401886 - SRR24401893
Reference genome or transcriptome used by the authors	<i>Chara braunii</i> genome
Main RNA-seq software or tools reported by the authors	cutadapt, STAR, DESeq2, and goseq

Part 4. Prepare a Simple RNA-Seq Experimental Design Diagram



- **What was changed or manipulated?**
 - The salinity level was manipulated by exposing *Chara braunii* to 5 practical salt units (PSU) salt stress for 6, 24, or 48 hours.
- **What was the control?**

- The control was *C. braunii* maintained in freshwater at 0 PSU. The medium was replaced with fresh deionized water without added salt, providing the 0-hour/0-PSU reference condition.
- **What was measured?**
 - The researchers measured physiological responses and changes in gene expression (transcriptome) after salt stress. For the RNA-seq part, they measured gene expression/transcript levels.
- **How many biological replicates were used? What tissue or cell type was sequenced?**
 - For the RNA-seq experiment, 2 biological replicates (duplicates) were used for each sample. The sequenced material was subsurface thalli segments of *C. braunii*.

Part 5. Locate the Public RNA-Seq Data

SRX	Condition	SRR Run Accession	Control/Treatment	# of Spots	# of Bases	Approx. Size
SRX20189891	Salt stress t24_1f	SRR24401886	Treatment	72M	14.5G	4.2Gb
SRX20189890	Salt stress t6_2f	SRR24401887	Treatment	61.6M	12.4G	3.6Gb
SRX20189889	Salt stress t6_1f	SRR24401888	Treatment	56.3M	11.4G	3.3Gb
SRX20189888	Salt stress t0_2f	SRR24401889	Control	57.1M	11.5G	3.4Gb
SRX20189887	Salt stress t48_2f	SRR24401890	Treatment	48.9M	9.9G	2.9Gb

Part 6. Assign One RNA-Seq Sample to Each Student

Run Accession Number	Condition	Biological Replicate Number	Single-end or Paired-end	Approximate File Size
SRR24401890	Treatment	Rep 1	Paired-end	2.9 Gb

Part 7. Download the RNA-Seq Data into Galaxy

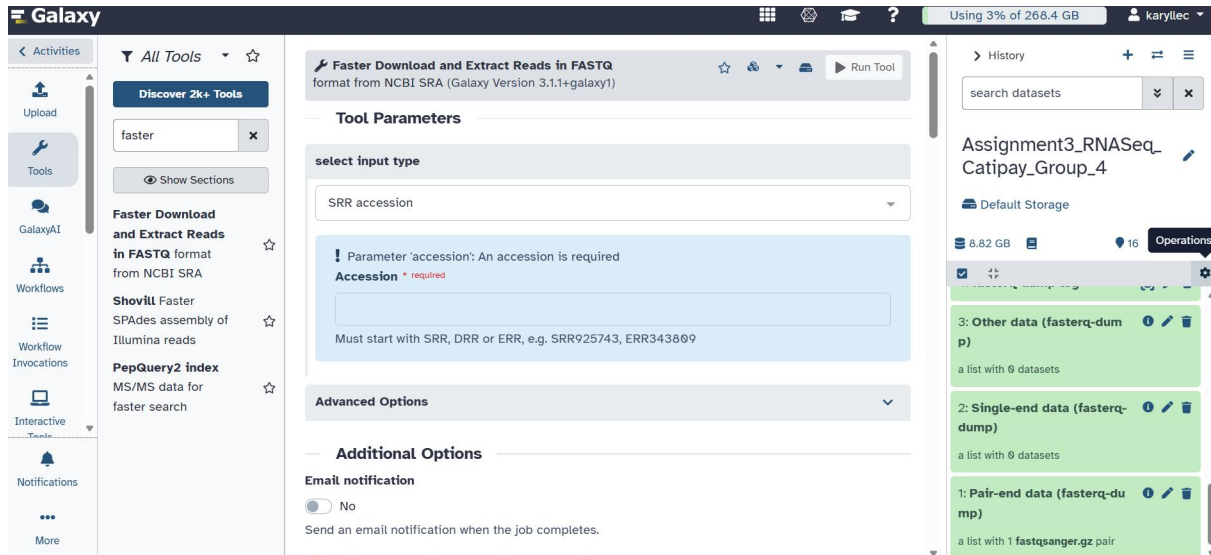


Figure 2. Successful download and extract of Pair_End and Single-End RNA Seq Reads in Galaxy using the tool FASTER Download and Extract Reads in FASTAQ.

Part 8. Examine the FASTQ File

Matrix	Control_Rep_R1	Control_Rep1_2
Line 1: sequence identifier	@SRR24401886.1/1	@SRR24401886.1/2
Line 2: nucleotide sequence	CAATCGTCCGCCACCGC TCGCTTTATAATCCTCC GCCACCGCCTCACGCAA ACACGCAGCTTTCCTT ACTCATCCCCCTTCGTG TTGTGACCACCTGAAA	ATCATCAAGTCATTGTG TCATACCTGGAAACCCA TCTCCTTCTGTCCCTGG TTAGTCGTGAGGACTTG GTTTTCAGGTGGTCACA ACACGAAGGGGGATGA
Line 3: separator line	+	+

Line 4: quality-score characters	<pre> FFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFF:FFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFF </pre>	<pre> FFFFFFFFF:FFF:F,FFFFFF FFFFFFFFFFFFFFFFFFFFFFF,FF FFFFFFFF,FFFF,FFFFFF,F,F ,F:FF:,FFFF:FFFF,FFFFFF FFFFF,FFFFF </pre>
----------------------------------	---	---

- How is a FASTQ file different from the FASTA genome file used in the previous assignments?
 - A FASTQ file contains nucleotide sequences together with their quality scores, while a FASTA file mainly contains the sequence and its identifier.
- What information is present in FASTQ that is not present in FASTA?
 - FASTQ contains quality-score characters for each nucleotide sequence. In your table, these are shown in Line 4 as characters such as F, :, ;, and ,.
- Why are quality scores important in RNA-seq?
 - Quality scores are important because they indicate the reliability of the sequencing reads. They help identify whether the bases in the RNA-seq data are high or low quality, which is important for obtaining reliable results during later analysis.

The screenshot shows the Galaxy web interface. The main panel displays the dataset '22: Treatment_T48_R2' with a size of 2.3 GB and format 'fastqsanger.gz'. A warning message indicates that only the first 100 KB is shown. The dataset content is displayed as follows:

```

@SRR24401890.1/2
GTCGTCCCATGCTTCTGTAGTTTGACCTACAGATGTCATATCACTATTGTTTGTGCTCTTGAACGACAACCTTGGTGC
TTGGCTCTGGAGGCCTGT
+
FFFFFFFF:FFFFFFFF:FFFFFFFF:FFFFFFFF:FFFFFFFF:FFFFFFFF:FFFFFFFF:FFFFFFFF:FFFFFFFF:
FFFFF:FFFFF::F::FF
@SRR24401890.2/2
GTGTATTATGTATTATGTAACATAACACTTTACGTACTCTCACCTTCCCTCAGGTTTAACTACTCAC
GCCTTCATCTCTTCTCT
+
FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF
FFFFFFFF:FFFFFFFF
@SRR24401890.3/2
CTGCACGACCGTCTGCTCTCTCTGAGCGACTAACAGCTAAACCATCTCTCGATCTCTCTCCCTCTCTCTGCTCTCT
TGCTGTCTTGCACGAC
+
:FFFFFFFFFFFFFFFF:FF,FFF:FFFF::F:FFFFFFFFFFFF:FF,F:FF:FFFFFF,FFFF,FFFF,FFFF::,
FFFF FFF ..FFFFFFF

```

Figure 3. Preview of the forward reads from paired-end RNA sequencing data in FASTQ format, generated using the Galaxy tool Unzip Collection to Forward Datasets.

Figure 4. Preview of the reverse reads from paired-end RNA sequencing data in FASTQ format, generated using the Galaxy tool Unzip Collection to Reverse Datasets.

Part 9. Run FastQC

Measure	Value
Filename	Treatment_T48_R2.gz
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9

Produced by [FastQC](#) (version 0.12.1)

Figure 5. FastQC Quality Control for Forward Read generated using the FastQC tool in Galaxy.

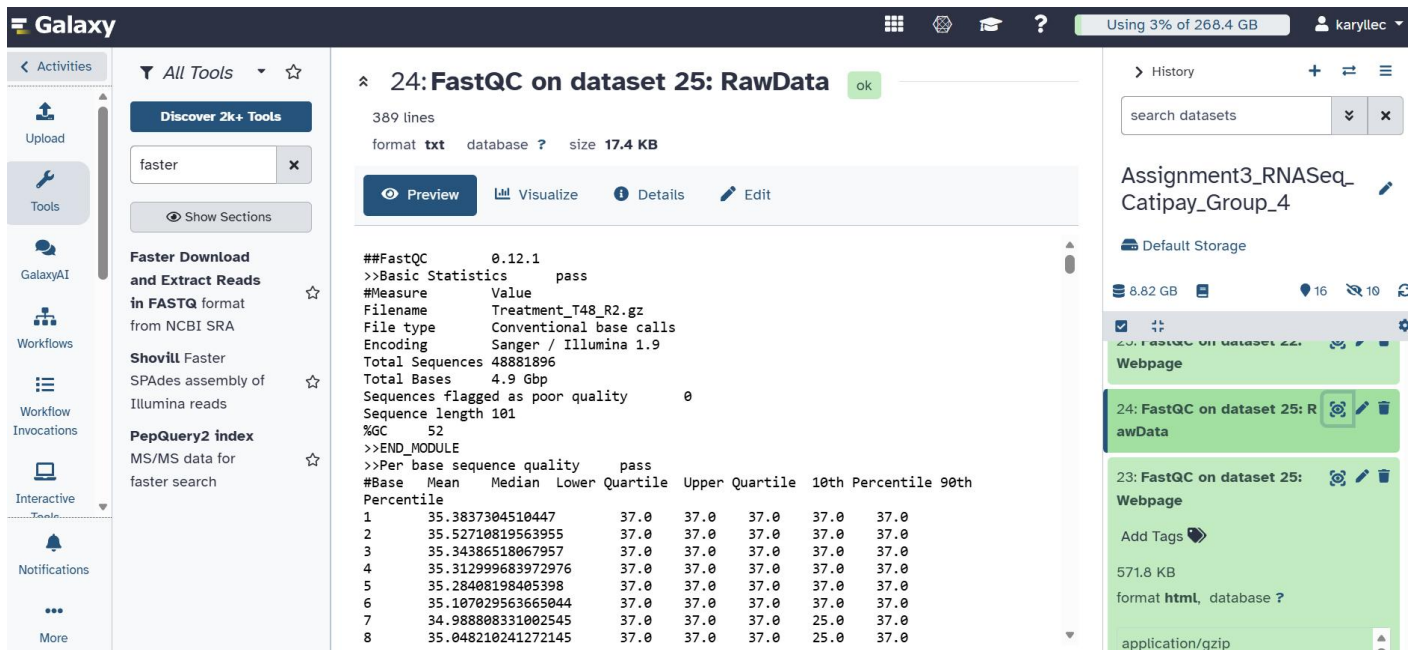


Figure 6. Raw Data Output of the FastQC Analysis for the Forward Read Generated Using the FastQC Tool in Galaxy

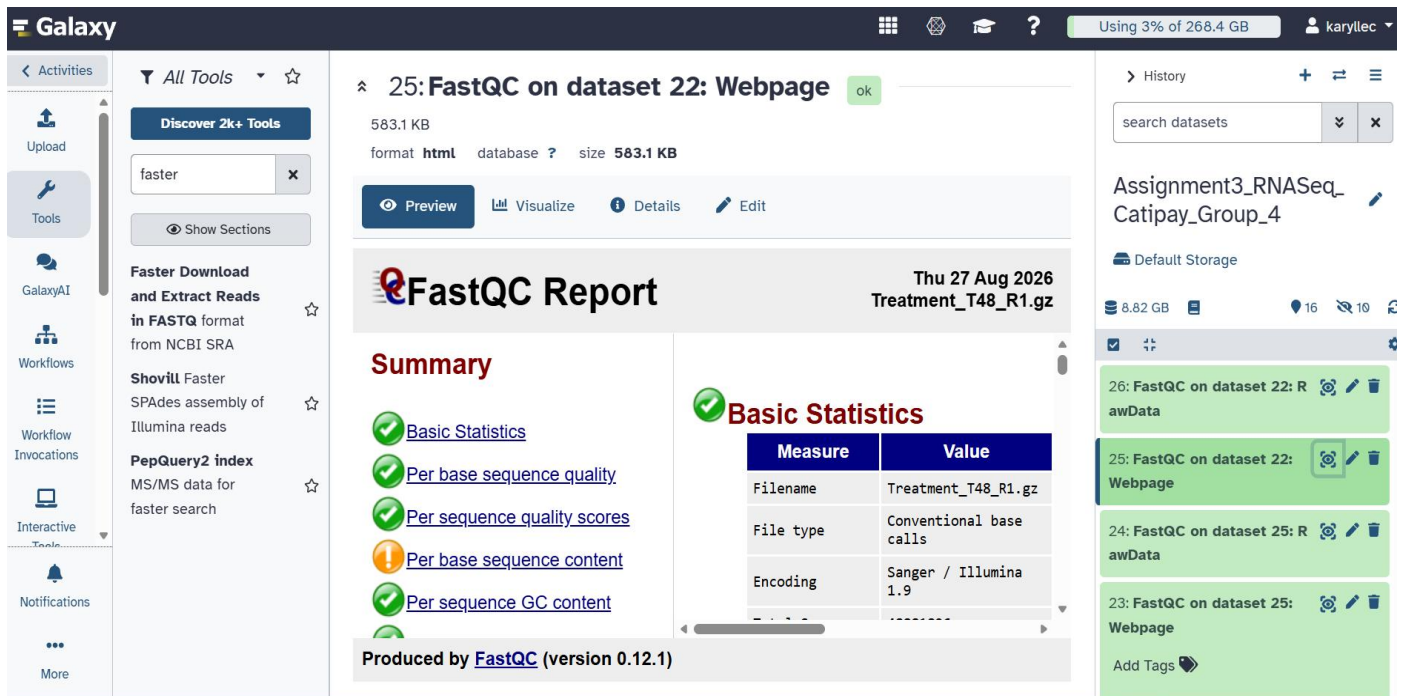


Figure 7. FastQC Quality Control Report for the Reverse Read Generated Using the FastQC Tool in Galaxy

Galaxy

Using 3% of 268.4 GB

karyillec

Activities

Upload

Tools

GalaxyAI

Workflows

Workflow Invocations

Interactive

Notifications

More

All Tools

Discover 2k+ Tools

faster

Show Sections

Faster Download and Extract Reads in FASTQ format from NCBI SRA

Shovill Faster SPAdes assembly of Illumina reads

PepQuery2 index MS/MS data for faster search

26: FastQC on dataset 22: RawData

386 lines

format txt database ? size 16.8 KB

Preview Visualize Details Edit

```
##FastQC 0.12.1
>>Basic Statistics pass
#Measure Value
Filename Treatment_T48_R1.gz
File type Conventional base calls
Encoding Sanger / Illumina 1.9
Total Sequences 48881896
Total Bases 4.9 Gbp
Sequences flagged as poor quality 0
Sequence length 101
%GC 51
>>END_MODULE
>>Per base sequence quality pass
#Base Mean Median Lower Quartile Upper Quartile 10th Percentile 90th Percentile
1 36.529470869951524 37.0 37.0 37.0 37.0 37.0
2 36.41676317137944 37.0 37.0 37.0 37.0 37.0
3 36.5468659808122 37.0 37.0 37.0 37.0 37.0
4 36.5422817887424 37.0 37.0 37.0 37.0 37.0
5 36.564478718255934 37.0 37.0 37.0 37.0 37.0
6 36.536794767535206 37.0 37.0 37.0 37.0 37.0
7 36.48791957660562 37.0 37.0 37.0 37.0 37.0
8 36.54190005232203 37.0 37.0 37.0 37.0 37.0
```

History

search datasets

Assignment3_RNASeq_Catipay_Group_4

Default Storage

8.82 GB 16 10

26: FastQC on dataset 22: RawData

25: FastQC on dataset 22: Webpage

24: FastQC on dataset 25: RawData

23: FastQC on dataset 25: Webpage

Add Tags

Figure 8. Raw Data Output of the FastQC Analysis for the Reverse Read Generated Using the FastQC Tool in Galaxy

Part 10. Characterize Your RNA-Seq Dataset

RNA-seq run accession	SRR24401890
Condition	Treatment – salt stress, 48 h (t48), Rep1
Single-end or paired-end	Paired-end
Number of reads / sequences	R1: 48,881,896 R2: 48,881,896
Read length	R1: 101 bp R2: 101 bp
File size	R1: 2.1 GB R2: 2.3 GB
GC content	R1: 51% R2: 52%
General per-base sequence quality	R1: Pass R2: Pass
Presence of adapter content	R1: Pass R2: Warn
Presence of overrepresented sequences	R1: Pass R2: Fail
Any quality warning observed	R1: Per base sequence content (warn), Sequence Duplication Levels (fail) R2: Per sequence GC content (fail), Per base sequence content (fail), Sequence Duplication Levels (fail), Overrepresented sequences (fail), Adapter Content (warn)

Part 12. Compare the RNA-Seq Samples Within the Group

As a group, combine the observations from all members. Prepare a small comparison table.

Student	Accession	Condition	Reads	Read length	GC %	Main QC observation
Balana, Jane Kate R.	SRR2440188 6	Treatment – salt stress, 24 h (t24)	72,004,032	101 bp	50%	PASS – Good overall sequencing quality with a high number of reads, 101 bp per-mate read length, and a balanced GC content of 50%. No major quality issues are indicated.
	87					
Trayvilla, Shekhan Fredianne C.	SRR2440188 8	Treatment - salt stress, 24 h (t24)	56,332,223	101 bp	50%	PASS – Excellent overall sequencing quality with high per-base quality (mean Q $\approx$ 34–36), no significant N content, and consistent 101 bp reads. However, poly-G sequences, high duplication, and biased per-base sequence composition were detected.
Olaguir, Junavhel Jane	SRR2440188 9	Control - freshwater (t0), Rep2	57, 100, 000	101 bp	51%	PASS – Good overall quality. Per-base quality is high (Pass). No adapter contamination. Some warnings for sequence duplication and overrepresented sequences, which is common in RNA-Seq.
Catipay, Nina Karylle	SRR2440189 0	Treatment - salt stress, 48 h (t48)	48,881,896	101 bp	51%	WARNING – Adapter content detected (Warn). Sequence duplication levels failed. Per-base sequence content failed. This sample has more

						quality issues than others, likely due to library prep or high expression of specific genes.
--	--	--	--	--	--	--

Part 13. Interpretation Questions

- What biological question was the original RNA-seq study trying to answer?
 - The study wanted to determine how *Chara braunii* responds and acclimates to salt stress at the physiological and gene-expression levels. The researchers exposed the algae to 5 PSU salt stress for 6, 24, and 48 hours and used RNA-seq to identify changes in gene expression. They were particularly interested in which genes were activated or suppressed during salt acclimation and what biological processes were involved.
- Why did the authors use RNA-seq instead of only examining the genome?
 - The genome shows what genes an organism has, but it does not tell us which genes are actively being expressed under a particular condition. RNA-seq allowed the researchers to examine the transcripts (mRNA) being produced by *C. braunii* during salt stress. This helped them identify genes whose expression increased or decreased after exposure to 5 PSU salt. The study quantified transcript levels for 17,387 genes and identified genes involved in processes such as ion/solute transport and cell-wall modification.
- What is the difference between genomic DNA and the RNA molecules measured by RNA-seq?
 - Genomic DNA is the organism's relatively stable genetic information and contains the genes and regulatory regions that make up its genome. RNA molecules, particularly mRNA, are produced when genes are expressed. Therefore, genomic DNA tells us what genes are present, while RNA-seq measures which genes are being expressed and how much transcript is being produced at a particular time and condition.

In this study, the researchers extracted total RNA from *C. braunii* thalli, isolated the mRNA, converted it into cDNA libraries, and sequenced those molecules.

- What is a biological replicate and why is it important?
 - A biological replicate is an independent biological sample that represents the same experimental condition. It is important because biological organisms naturally vary, so using replicates helps researchers determine whether an observed difference is due to the treatment rather than random biological variation.

In this study, the RNA samples were prepared in biological duplicates for each sample, meaning each RNA-seq condition had two independent biological replicates. The researchers could then compare the replicates to assess whether their gene-expression responses were consistent.

5. What is the difference between single-end and paired-end sequencing?

- In single-end sequencing, each RNA-seq fragment is sequenced from one end, producing one sequence read per fragment.

In paired-end sequencing, each fragment is sequenced from both ends, producing two reads: R1 (forward) and R2 (reverse). These two reads correspond to the same original fragment and provide more information about its sequence and location.

6. What is a FASTQ file?

- A FASTQ file is a text-based format used in bioinformatics for storing both a biological sequence and its corresponding base quality scores. Each sequence entry typically consists of four lines: a sequence identifier header starting with @, the raw sequence string, a separator header starting with +, and quality score characters for each base.

7. What information does FastQC provide?

- FastQC is a quality control tool that analyzes raw high-throughput sequencing data and generates a summary report. It provides metrics such as, per-base and per-sequence quality scores, per-base sequence content, per-sequence GC content distribution, per-base N content, sequence length distribution, sequence duplication levels and overrepresented sequences, and possible adapter content.

8. What does a high per-base quality score indicate?

- A high per-base quality score indicates high confidence in the accuracy of the base call. It means the probability that a base was called incorrectly by the sequencing instrument is extremely low.

9. Why can adapter contamination be a problem?

- Adapter contamination happens when sequencing reads accidentally include artificial laboratory adapters. Because these extra sequences do not belong to the organism, they confuse computers during analysis, leading to mapping errors and incorrect gene measurement. Trimming these synthetic pieces out ensures the final data is accurate and clean.

10. Were all RNA-seq samples in your group similar in quality? Explain.

- Mostly yes, but not entirely. Most samples passed the per-base quality and adapter checks. However, some samples (like SRR24401890) showed warnings for adapter content and sequence duplication, while others had more severe failures in duplication levels. This suggests some variation in library preparation or biological complexity.

11. Did any sample show a possible quality problem? What was it?

- Yes. SRR24401890 showed a "Warn" for Adapter Content and "Fail" for Sequence Duplication Levels. SRR24401891 (Itom's sample) also had multiple "Fail" warnings for duplication and overrepresented sequences. These issues may require trimming or filtering before analysis.

12. What additional steps would be needed before the researchers could compare gene expression between control and treatment samples?

- To compare gene expression between control and treatment samples, researchers must first trim adapter sequences and filter out low-quality bases using Cutadapt. Next, the cleaned reads are aligned to the *Chara braunii* reference genome using STAR. The mapped reads are then quantified for each gene to build a raw expression count matrix. Finally, this data is normalized and statistically analyzed using DESeq2 to identify significantly differentially expressed genes.

Part 15. GROUP SUMMARY

This activity evaluated the RNA-Seq data from *Chara braunii* under salt stress (5 PSU) across multiple exposure times using tools in Galaxy. The quality control assessment showed that while most samples exhibited high per-base sequence quality and consistent GC content, several datasets (such as SRR24401890) flagged warnings or failures due to sequence duplication, adapter content, and overrepresented sequences. These findings highlight the importance of pre-processing steps, such as adapter trimming and duplicate removal, prior to downstream analysis. Ultimately, quality-checked raw reads are essential to ensure accurate alignment and reliable differential gene expression profiling during salt acclimation.

Article Link

Heß, D., Heise, C.M., Schubert, H., Hess, W.R. & Hagemann, M. (2023) *The impact of salt stress on the physiology and the transcriptome of the model streptophyte green alga Chara braunii*. *Physiologia Plantarum*, 175(6), e14123. Available from: <https://onlinelibrary.wiley.com/doi/10.1111/ppl.14123>